EMBA 756: MANAGING OPERATIONS

LEARNING JOURNAL FOR "LEAN ANTHOLOGY"

The purpose of this assignment is for you to apply what you read through the book of **"Lean Anthology"** as you go through our "Managing Operations" course. As the book says in its introduction, "give some thoughts to any parallels you notice in your own life, whether on a personal or professional level." We would love to see you discover new ways of improving your personal and professional processes as a result of the course and reading this book.

Our "learning journal" will get a little more specific, as outlined in the described tasks below, but it will leave enough flexibility for you to choose the environments for which you choose to apply the learning.

Specifics of the Assignment: As an **individual write-up,** you have to provide answers for the **first three TASKS, and for one of the other two remaining tasks.** In total you are responsible for four out of the five tasks**.** All five tasks are described below. For each task, you have a **choice** between answering the **"original task"** or **one of its variations**, as you find it more appropriate for your learning. I expect no more than 5 pages and relevant exhibits. (It is a guideline, not a page limit).

Think of the "learning journal" as your **midterm individual assignment.**

DUE DATE: Two weeks after our first weekend, specifically June 16 by 5 pm. Send your file (prefer Word file or PDF) as an attachment to kouvelis@wustl.edu.

Grading of the assignment: It will be graded on effort, level of learning, and effective communication. It will receive one of three grades: **Excellent**, **Enough to pass**, and **Lacking Effort**. The "lacking effort" may contribute to a LP for the course.

TASK 1: Read Chapter 1, "Defining the Customer Value Proposition."

Original Task: We need to first understand how processes support their customers (external or internal). What are the Order Winning Criteria? Do your processes have the capabilities to support them? Or, how are you going to build these capabilities into your processes after you choose the OWC? What happens when there are heterogeneous market segment with very different OWCs that cannot be supported through a single process?

Well, time for you to apply these concepts. Describe a business process of yours, Voice of Its Customer(s) (customers can be external or internal), and the relevant OWCs the process is supporting (or you are planning to support with appropriate process redesign).

Variation of Task 1.1: If you do not have a business process of your own to describe, then apply it at the personal level as the “Brain Play” question 3 on page 10 of the book asks you to.

TASK 2: Read **Chapter** 2, “Managing Variability”

Original Task: Managing Variability (internal, as a result of operating variables, or external, as a result of customers and or suppliers) is the challenge for all processes. It becomes especially important during the peak time of our operations. We place a lot of emphasis on it in our course, and our “Littlefield Labs” simulation.

Well, time for you to apply these concepts in your own business processes. Describe for a business process of yours on how you use inventory, capacity or lead-time to deal with unexpected variability faced by your processes. Provide enough details to understand the nature of your process, the nature of the variability you face, and how the three operating levers are used (or you are planning to use them) to effectively deal with variability in support of your pursued customer value proposition (support of your OWC).

Variation of Task 2.1: If you cannot apply it to a business process of yours, then answer “Brain Play” question 3 on page 21 of the book.

Variation of Task 2.2: Start by answering question 2 in the “Brain Play” part of this chapter, on page 21. Also describe differences between different fast food restaurants you might have visited on how they differ in their approaches in managing variability (if any). At least in my mind, Chipotle and MacDonald’s are very different in the way they are managing variability (as a hint on where to look for differences).Of course, the Waffle House is a class of itself (and very close to my heart).

TASK 3: Read chapter 3, “Understanding Little’s Law”

Original Task: Now we start looking at our processes as “flow processes.” Something flows through them (customers, orders, products, money etc.). We are interested within our defined “box” how many flow units are there (inventory), how long it takes to go from the beggining of our “box” to the end of it (throughput time or flow time), and how many units get through the box per unit time (throughput rate or flow rate). That is the language of operations. And Little’s Law is the most important one describing the behavior of flow processes:

Inventory = Throughput Rate (times) Throughput (Flow) Time.

Now time for you to describe your business process that way. What flows through it, what is the inventory, what is the throughput rate, and what is the throughput time. Reminder: All the measures used in Little's Law are time averages over long periods of time, and with operations being at "steady state."

Variation of Task 3.1: Start by answering "Brain Play" question 1 on page 33. Continue then to explain how you will apply Little's Law in an HR type process, like "hiring for a specific position."

Variation of Task 3.2: Answer "Brain Play" question 2 on page 33. Then answer question 3 in the "Brain Play" portion on page 33, but with focus on a situation from your personal life.

Chapters 4 and 5 in the book are excellent readings on the importance of small batch sizes and the overall logic of the theory of constraints (managing the system through its "bottleneck resources"). You get enough exposure and practice to these concepts in the book the GOAL and its book report, that I will skip any learning task here.

Chapters 5 through 9 get into designing, developing and perfecting the processes that will support our value proposition to our "customers" and deliver our OWC. That is where you see some of the Lean philosophy and tools. Interesting readings with direct application to everybody's personal life, and I hope you get inspired to use them to improve your personal lives.

From a course learning perspective, I want us to practice more the **VALUE STREAM MAPPING** tool. And I expect that all of you have applied at some point in time in your life (not necessarily for this journal) the "standard work" **5S** tools logic.

TASK 4: Think about a business process that you would like to improve. First, Value stream map (VSM) its current state. Identify and remove "muda" (see page 90 of the book). Then, think the next target state you would like to move the process to, i.e. VSM its future state. Describe implementation steps for going from current state to future state, and how this transformation helps achieving the top line metrics for your process and supporting your OWC.

Variation of task 4.1: How could an airline or an insurance company that is focused on improving turnaround times use some of these techniques (value stream mapping, critical path, standard work etc.)?

Variation of task 4.2: How could a try cleaning establishment improve their functional space using 5S tools? Offer a few specific ideas using your local dry cleaner as an example.

We are now going to shift our attention to Chapters 13 through 15 (the so called "Coordination Part," but really more about logic of inventories and their management for operations and supply chains).

TASK 5: Think about an inventory situation either at work or at home.

(**Task Variation:** If you cannot find any such example, then think about inventories at convenience store places like Seven Eleven or the non-drug related part of Walgreens. Well, if you are more into electronics, then think about a large electronics retailer, like Best Buy).

5a): What inventory system, continuous review or periodic review might be appropriate? If there are multiple inventoried items, try to differentiate between items in the use of such systems. Explain the logic of your answer.

5b): Is it possible to implement a visual management system for your inventory control and reordering of some of the items discussed in 5a? How would you design and implement your system? What would you measure as appropriate performance measures of it?

Well, we are now at the end of our assignment on applying lean to your context. Hope, you enjoyed it. Three quick points of advice:

1) Do not let perfect get in the way of better;

2) Lean is the only race you lose by finishing. It is the relentless pursuit that matters.

3) Lean is about generating the most value out of your effort. It all about effectiveness.